

Claire AI Execution Flow Specification

****CLAIRE AI -- EXECUTION FLOW SPECIFICATION****

End-to-End System Behavior Medium-Wide Layout Unicode Symbols Only Export-Ready

1. Overview

This document defines the complete execution flow of Claire AI across all modes:

Deterministic Mode

Connected Intelligence Mode

Hybrid Mode

It describes:

data flow

engine activation

safety gating

breakthrough detection

scoring

promotion

portfolio creation

clustering

dashboard output

This is the operational blueprint of the entire system.

2. High-Level Execution Flow (Unicode Diagram)

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CLAIRE AI -- EXECUTION FLOW User Input / External Signals v Mode-Switching
Controller v Engine Activation (Deterministic / Connected / Hybrid) v
Semantic Processing Layer v Breakthrough Detection Engine v Scoring &
Validation Layer v Portfolio Candidate Object (PCO) v Portfolio Creator
Engine v Portfolio Engine (Clusters, Themes, Adjacencies) v Portfolio
Dashboard Output Safety & Governance Layer (Overrides at Every Stage)
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3. Mode-Dependent Execution Behavior

3.1 Deterministic Mode

No external ingestion

Pure internal logic

Fully reproducible

Patent-safe

Breakthroughs generated from semantic + concept graph reasoning

Portfolio creation allowed

Safety layer enforces isolation

3.2 Connected Intelligence Mode

Live ingestion active

Real-time signals processed

Breakthroughs influenced by external data

Portfolio creation allowed

Safety layer governs ingestion and redline triggers

3.3 Hybrid Mode

Both engines active

Dual-pipeline fusion

Highest-fidelity breakthroughs

Portfolio creation allowed

Safety layer governs dual-pipeline interactions

4. Detailed Execution Flow (Step-by-Step)

Below is the full end-to-end execution flow, broken into 12 stages.

Stage 1 -- Mode Initialization

Claire determines active mode:

deterministic

connected

hybrid

Mode-Switching Controller:

activates engines

enforces isolation
checks safety matrix
logs mode state

Stage 2 -- Data Ingestion (Connected & Hybrid Only)

If ingestion is active:

tech signals
market signals
competitor signals
regulatory signals flow into the Ingestion Layer.

Safety Layer:

filters
blocks
downgrades
logs any redline content.

Stage 3 -- Semantic Processing Layer

All modes use this layer. It performs:

semantic extraction
concept graph expansion
deterministic embeddings
live semantic overlays (if connected/hybrid)

Output:

semantic_vectors
concept_graph
feature_maps

Stage 4 -- Hybrid Fusion (Hybrid Mode Only)

If hybrid mode is active:

deterministic semantic maps
live semantic overlays are merged into a hybrid intelligence substrate.

This produces:

fused novelty vectors
fused feasibility vectors
fused relevance vectors

Stage 5 -- Breakthrough Detection Engine

This engine analyzes:

semantic structures

contradictions

missing functions

cross-domain opportunities

live signals (if available)

hybrid fused vectors (if hybrid)

Output:

Breakthrough Object (BO)

Stage 6 -- Safety Pre-Validation

Safety Layer evaluates the BO:

export-control risk

dual-use risk

classified-adjacent risk

redline triggers

compliance state

If unsafe:

BO is blocked

pipeline halts

governance review required

Stage 7 -- Scoring & Validation Layer

If BO passes safety:

novelty scoring

feasibility scoring

market alignment scoring

timing scoring

competitive pressure scoring

acquirer fit scoring

risk scoring

Composite score is generated.

If composite score threshold:

BO -> PCO (Portfolio Candidate Object)

If not:

BO archived

Stage 8 -- Portfolio Candidate Object (PCO) Creation

PCO includes:

rationale

scores

safety flags

governance requirements

If safety flags exist:

PCO enters "pending approval"

If clean:

PCO auto-promotes

Stage 9 -- Portfolio Creator Engine

This engine transforms the PCO into a Portfolio Item (PI). It generates:

****Technical Package****

architecture

components

feasibility model

****Market Package****

TAM/SAM/SOM

competitive landscape

adoption timing

****Strategic Package****

acquirer fit

capability gap map

strategic timing

****Risk Package****

technical

market

regulatory

Stage 10 -- Portfolio Engine

Portfolio Items are:

clustered
themed
adjacency-mapped
ranked

This creates the portfolio structure.

Stage 11 -- Portfolio Dashboard

Outputs:

new items
updated clusters
opportunity windows
acquirer matches
risk profiles
strategic insights

Stage 12 -- Safety & Governance Oversight

At every stage, Safety Layer can:

block
downgrade
isolate
require approval
revert to deterministic mode

This ensures federal-grade compliance.

5. Execution Flow Summary Table

Stage	Component	Function
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1	Mode Controller	Activates Engines, Enforces Isolation
2	Ingestion Layer	Processes External Signals
3	Semantic Layer	Extracts Meaning, Builds graphs

4	Hybrid Fusion	Merges Deterministic+ Live Intelligence
5	Breakthrough Engine	Generates BO
6	Safety Pre-Validation	Blocks Unsafe Breakthroughs
7	Scoring Layer	Validates and scores BO
8	PCO Creation	Promotes Breakthrough
9	Portfolio Creator	Generates PI
10	Portfolio Engine	Clusters and Themes
11	Dashboard	Outputs Insights
12	Safety Layer	Overrides at all stages

6. Summary

This execution flow specification:
defines Claire's complete operational behavior
shows how engines interact
demonstrates safety enforcement
proves autonomous pipeline logic
satisfies acquisition-grade diligence